

Distribution Kinetics Modeling of Nucleation, Growth, and Aggregation Processes

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Certain emerging fields of nanotechnology depend on self-assembly processes that lead to molecular structures which are organized and ordered or disorganized and disordered. Many such processes can be understood as combinations of nucleation, reversible growth, and reversible aggregation. We develop a descriptive theoretical framework through population balances for the cluster mass distribution and show the formulation of its mass moments. Calculations for relaxation to equilibrium show the effects of aggregation, deaggregation, dissolution, cluster growth by monomer addition, and nucleation, as either single processes or various combinations. Competition among the processes yields a variety of different behaviors that have been observed in nature and in the laboratory. Problems that can hinder self-organization processes include coarsening (Ostwald ripening) and glasslike phase transition (noncrystalline solidification). Both of these and other processes can be described by the kinetics of cluster distributions and the dynamics of their moments.

Introduction

Understanding the dynamics of cluster nucleation, growth and dissolution, and aggregation–deaggregation is important in many scientific fields, including geology, biology, chemistry, medicine, and chemical engineering.^{1–3} An important issue for the kinetics and dynamics of self-organizing processes is the competition between nucleation, growth, and aggregation. Nucleation combined with growth to produce observable crystalline structures may require considerable time compared to aggregation of small crystals into random clusters. Systems with relatively slow aggregation rates will therefore allow growth of the preferred crystalline structures. Conversely, systems that aggregate relatively rapidly may yield formless aggregates with glass properties. Glassy aggregates are characterized by a large value of viscosity, implying large, densely packed clusters.

Much effort has been directed toward mathematical modeling of the *separate* processes of nucleation, growth, and agglomeration for physical and chemical reaction processes. Only slight progress has been made in combining these phenomena to assess the effects of their competitive interactions. The methods of *distribution kinetics* allow systematic mathematical modeling that leads to the formulation and solution of governing equations for these *reversible* processes. We assume nucleation follows classical homogeneous theory or is caused by heterogeneous nuclei added to the solution. Growth due to monomer addition from solution to clusters and aggregation between clusters are both represented by integrals of the cluster distribution. When growth and aggregation rate coefficients are independent of or weakly dependent on cluster size, the population balance equations are readily solved by moment analysis. By incorporation of solute (monomer) depletion, the results afford realistic behavior for the cluster number and mass concentration. The monomer

addition terms are shown to be consistent with (and a generalization of) conventional differential growth and growth dispersion expressions. Compared with the irreversible differential approach, the generalization to reversible growth and aggregation reveals differences that are rich in possibilities for explaining particulate dynamics. The present objective is to demonstrate how distribution kinetics can be applied to develop population balance equations that describe nucleation, growth, and aggregation. The moment solutions to these equations for open (flow) and closed (batch) systems illustrate the general dynamic behavior. A goal is to identify key equations and parameters and to explore the major behavior of such systems. The theories will be verified by comparison to the general observed behavior for precipitation, crystal growth, Ostwald ripening, glass transition, and nucleation induction. A result of this work is the realization that a single consistent and coherent framework, based on chemical reaction principles, has the potential to characterize a wide range of particulate reactions and phase transformations. Developing such methods is vital to bridging the gap between conventional chemical engineering and advanced materials science.

Typically, crystals or aggregates have nanoscale dimensions and are distributed in size depending on their history. Precipitation of particulates from solution yields a *cluster size distribution* (CSD) that tends to evolve with time until an equilibrium or steady-state condition is attained. The CSD is shaped by the fundamental thermodynamic, transport, and chemical kinetics processes underlying the evolution. Evolving populations of entities distributed in some property are commonly modeled⁴ by population balance equations—a method for quantitatively describing the distribution dynamics through mass balances expressed as equations for the CSD. Although much is understood about CSD dynamics, critical gaps in knowledge continue to motivate investigations of cluster dynamics and aggregation. Self-assembly can occur by a nucleation mechanism with subsequent growth (or crystallization) of building

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units, followed by aggregation (assembly). Possible benefits of nanoscale studies include a better understanding of molecular processes in the environment and the development of manufacturing processes that will yield valuable products.

Nielsen's monograph⁵ reviewed the kinetics of precipitation, focusing on crystals in liquid solutions. Following the numerical approach used by Hounslow,⁶ Chakraborty and Bhatia⁷ discussed the modeling of precipitation and aggregation of inorganic crystals. They indicated that urinary stones owe their formation to agglomeration of calcium oxalate, rather than particle growth only. Most engineering studies treat agglomeration or aggregation as hindrances to crystal manufacture, but a modern outlook is that such phenomena can be channeled or controlled to yield nanosystems of high value. Hounslow and co-workers^{8,9} have treated crystal precipitation processes involving nucleation, growth, and aggregation. These approaches are inherently irreversible and require heuristic addition of arbitrary terms to describe growth dispersion. Our treatment of vapor-liquid nucleation and growth is intrinsically reversible,¹⁰ describing either growth or evaporation of aerosol droplets. The crystal growth theory proposed recently¹¹ for nucleation and growth is likewise reversible with crystal dissolution exhibited when supersaturation is less than unity. Owing to the foundation in polymerization-depolymerization kinetics, growth dispersion is incorporated naturally, as we will show below.

Self-assembly, a phenomenon that offers great promise for fabricating novel materials, can occur by a nucleation mechanism with subsequent growth (or crystallization) of building units and then aggregation (assembly). Crystal growth is the simplest instance of self-assembly and needs further study to explore its implications for more complex, ordered constructions. Mann et al.¹² proposed that biomineralization is a result of biochemically controlled nucleation and growth of inorganic crystals. In an elegant discussion of self-assembly in biomineral and biomimetic materials chemistry, Mann¹³ states that chemical fabrication of organized inorganic-based architectures is critically dependent on organic supramolecular assemblies. Several stages are envisioned in the fabrication: (i) organized supramolecular assemblies are formed, (ii) headgroups are spatially organized and chemically functionalized to form a template, (iii) oriented crystallization occurs, possibly over length scales that control vectorial growth, forming elaborate microstructures, (iv) networks of supramolecular assemblies are processed by cellular forces to construct higher-order architectures, often exhibiting hierarchical organization. Although complex, this scheme embodies phenomena that can be addressed by incorporating fundamental quantitative descriptions in the balance equations through the use of rate coefficients representing the different kinetic steps.

Zeolite molecular sieves, of immense importance in catalysis, separations, and ion exchange, can be manufactured by a hierarchical procedure.^{14,15} Mintova et al.¹⁶ described experiments for nucleation and growth of zeolite molecular sieves from clear solution at room temperature. Transmission electron microscopy showed that colloidal suspensions of aluminosilicate gel agglomerates are associated with nucleation and conversion into the final zeolite phase. The mechanism for assembly of the lattice through aggregation and realignment or preassembled building blocks on template

clusters remains uncertain. We suggest that distribution kinetics can shed light on the alternative possibilities for the mechanism, just as it allows discrimination among alternative mechanisms for free-radical polymer reactions.¹⁷ Thermodynamicists have led the discussions on research in nanotechnology, yet the metastable character and consequent dynamics of nanoparticles are effects of *kinetics*. Thermodynamic equilibrium is a limiting (long-time) case of the differential equations, which include the relevant underlying fundamental processes. Distribution kinetics preserves the minimum Gibbs free energy definition of the equilibrium state.¹⁸

A recent advance in aggregation-based crystal growth illustrates the importance of our proposed work. Alivasatos,¹⁹ in a perspective on Banfield et al.,²⁰ describes how traditional models attribute crystal growth to single molecule or atom addition. Banfield and co-workers,²⁰ however, have observed crystal growth by nanocrystal addition—a preformed modular-unit construction. The advantage of this crystal growth mode is that small nanocrystals need not contain interior defects (smaller crystals are usually more ordered than large crystals). The assembled aggregates built from perfect nanocrystals can yield highly ordered crystals as a result of ordered alignment, docking, and fusing steps. This self-assembly of oriented nanocrystals observed in the laboratory is ubiquitous in nature. We hypothesize that these coordinated steps can be represented by a judicious choice of rate coefficients in a population balance approach that distinguishes among the hierarchical structures.

The methods of *distribution kinetics* allow straightforward mathematical formulation and solution for combined reversible nucleation, growth, and agglomeration. Reiss²¹ proposed the analogy between chainwise polymerization and growth of clusters and how it could be applied. The approach has attractive features, as illustrated in the treatment of vapor-liquid nucleation and growth.¹⁰ The current work is based on a theory of *crystal* nucleation and growth,¹¹ which followed a series of papers on macromolecular processes.^{4,22,23} The aim is to demonstrate how distribution kinetics can be applied to develop population balance equations that describe nucleation, growth, and aggregation—all *reversible processes*. The moment solutions to these equations for open (flow) and closed (batch) systems illustrate the general mathematical behavior. The theory must necessarily describe phenomena occurring over multiple spatial and temporal scales. A goal is to identify key equations and parameters and to explore the simplest representations that yield nontrivial information. The change in monomer (substrate) concentration and its effect on particle size is treated rigorously.

Ortoleva²⁴ explained that competition between ordered crystallization and disordered clustering can lead to glass formation. Molecules can deposit on a crystal lattice either in an ordered manner compatible with the lattice or in a disoriented way that would require reorientation to achieve crystalline order. The reorientation process has a characteristic time (or rate coefficient) such that at low temperatures the probability of reorientation may be reduced. If the temperature is lowered sufficiently rapidly, few molecules or clusters will form ordered structures, but the viscosity will increase as clusters grow larger and merge, resulting in a glassy solid. Slow-temperature reduction, however, would allow nucleation and crystal growth, resulting in

a crystalline solid. Even after a glassy solid has formed, nucleation and crystal growth may continue over geologic time scales. According to Ortoleva,²⁴ "Natural glasses do not survive more than a few million years." Clustering is clearly an impediment to the development of crystalline order, and once a glassy state has been frozen in, annealing to the crystal state is extremely difficult.

The hypothesis that cooperative movements of clusters of particles can exhibit liquid–solid-phase transitions was investigated by Weeks et al.²⁵ for colloidal suspensions. They observed CSDs, structures, and dynamics of particles in colloidal supercooled fluids and colloidal glasses. Their observation of stringlike or extended fractal clusters, whose size increased as the glass transition was approached, is suggestive of polymers.

Many theoretical treatments of cluster dynamics are based on generalized thermodynamic models or computer experiments.² Our intent in this work is to explore the hypothesis that the competition among nucleation, reversible monomer addition (crystal growth), and reversible random clustering can be represented by population balance equations and their analytical moment equations. The governing equations are dynamic equations for cluster mass distributions that represent monomer addition and crystal dissolution and random aggregation and fragmentation (breakage) of clusters along with nucleation. This distribution kinetics approach provides algebraic and differential equations that not only are readily solved but have easily understood physical meaning at every step. In this initial attempt at building a kinetics foundation for liquid–glass–crystal-phase transitions, we utilize a simplified representation of processes and structures. For example, we assume that rate coefficients are independent of size but not of temperature.

Nucleation and Cluster Distributions

Homogeneous nucleation can be represented as growth by clustering of solute molecules, which approximate ions in stoichiometric proportion.²⁶ The free energy of formation, expressed in terms of interfacial energy σ , supersaturation S , molar volume V , Avogadro's number N_A , and a geometric factor ϵ , reaches a maximum

$$E^* = N_A \epsilon \sigma^2 V^3 / (RT \ln S)^2 \quad (1.1)$$

when the critical radius is

$$r^* = 2\sigma v / RT \ln S \quad (1.2)$$

Clusters smaller than r^* are unstable and tend to dissolve; clusters larger than r^* are stable and will grow spontaneously as long as $S > 1$. The rate of homogeneous nucleation, here written with a prefactor k^* , is the flux over the energy barrier

$$I = k^* \exp(-E^*/RT) \quad (1.3)$$

and increases strongly with S and T . Growth, like nucleation, also depends on S , an interrelationship that was emphasized in developments by Chiang and Donohue,²⁷ as well as by Mohanty et al.²⁶ These theories did not, however, apply population balance modeling as is done in the present work.

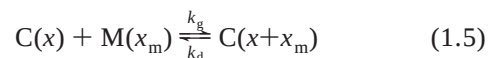
The cluster (or particle) mass distribution is defined so that $c(x,t) dx$ is the molar concentration of clusters

having values of mass x in the range x to $x + dx$. Integral forms of the rate expressions in the population balances lend themselves to calculations by moments, defined as integrals over x

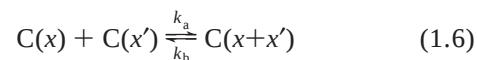
$$c^{(n)}(t) = \int_0^\infty c(x,t) x^n dx \quad (1.4)$$

The zeroth moment, $c^{(0)}(t)$, is the time-dependent molar concentration, and the first moment, $c^{(1)}(t)$, is the mass concentration (mass/volume) of clusters. The average cluster (particle) mass is the ratio $c^{\text{avg}} = c^{(1)}/c^{(0)}$. The molar concentration, $m^{(0)}(t)$, of monomers (substrate) of molecular weight x_m is the zeroth moment of the monomer distribution, $m(x,t) = m^{(0)}(t) \delta(x-x_m)$. The Dirac delta with specific mass, x_m , ensures that the theory has the correct semicontinuous character. An analysis of how continuous and discrete population balance models are related, and the advantages and disadvantages of each, was recently presented.²⁸

A general representation of reversible addition of monomer, $M(x_m)$, to a cluster, $C(x)$, is



The rate coefficients, k_g and k_d , are for growth and dissolution, respectively. The clusters may aggregate with rate coefficient k_a or fragment with breakage rate coefficient k_b



Mass conservation is ensured by the mass representation in eqs 1.5 and 1.6 and by exact formulation of population balance equations. To illustrate our distribution kinetics approach, we next present special cases of these general growth and aggregation processes. We assume in this work that the rate coefficients are independent of x , an approximation that allows realization of major effects.

Kinetics of Nucleation, Growth, and Aggregation

To derive the population balance equations, consider a well-mixed vessel of volume V with total flow rate Q and define the average residence time, $\tau_r = V/Q$. We consider the densities of liquid and clusters (or crystals) to be identical, so that the total volume V and flow rate Q (in and out) are constant. The inlet distributions for clusters and liquid are $c_i(x)$ and $m_i(x)$. Heterogeneous nucleation requires initial, $c_0(x)$, or input, $c_i(x)$, distributions (seeds). In the population balance, the homogeneous nucleation rate¹⁰ for clusters of mass x^* is $I\delta(x-x^*)$. Loss terms in eqs 1.5 and 1.6 have negative signs and generation terms have plus signs.^{4,22,23} Combination (addition) reactions are second-order rate expressions in the combining particle distributions, while fragmentation (dissociation) reactions are first-order.⁴ Fragmentation (eq 1.6) is assumed to be random so that products are uniformly distributed.²² The balance equations^{4,10} that govern the distributions of the clusters, $c(x,t)$, and

the monomer, $m(x,t) = m^{(0)}(t) \delta(x-x_m)$, are

$$\begin{aligned} \partial c / \partial t = & -k_g c \int_0^\infty m(x') dx' + k_g \int_0^{x_m} c(x-x') m(x') dx' - \\ & k_d c + k_d \int_x^\infty c(x') \delta[x-(x'-x_m)] dx' - 2k_a c \int_0^\infty c(x') dx' + \\ & k_a \int_0^{x_m} c(x') c(x-x') dx' - k_b c + 2k_b \int_x^\infty c(x') dx' / x' + \\ & I \delta(x-x^*) + (c_i - c) / \tau_r \quad (2.1) \end{aligned}$$

and

$$\begin{aligned} \partial m / \partial t = & -k_g m(x) \int_0^\infty c(x') dx' + k_d \int_x^\infty c(x') \times \\ & \delta(x-x_m) dx' - I \delta(x-x^*) x^* / x_m + (m_i - m) / \tau_r \quad (2.2) \end{aligned}$$

The general moment equations are determined²² by applying the operation $\int_0^\infty [] x^n dx$.

$$\begin{aligned} dc^{(n)} / dt = & -(k_d + k_g m^{(0)}) c^{(n)} + k_g \sum_{j=0}^n \binom{n}{j} x_m^{n-j} c^{(j)} \times \\ & [(-1)^{n-j} k_d + k_g m^{(0)}] - 2k_a c^{(0)} c^{(n)} + k_a \sum_{j=0}^n \binom{n}{j} c^{(j)} c^{(n-j)} - \\ & k_b c^{(n)} + 2k_b c^{(n)} / (n+1) + I x^{*n} + (c_i^{(n)} - c^{(n)}) / \tau_r \quad (2.3) \end{aligned}$$

and

$$dm^{(0)} / dt = [k_d - k_g m^{(0)}] c^{(0)} - I x^* / x_m + (m_i^{(0)} - m^{(0)}) / \tau_r \quad (2.4)$$

Details of the algebraic manipulations to attain these moment equations have been explained.¹⁰ The cluster moment equations for $n = 0, 1$, and 2 are

$$dc^{(0)} / dt = [k_b - k_a c^{(0)}] c^{(0)} + I + (c_{in}^{(0)} - c^{(0)}) / \tau_r \quad (2.5)$$

$$dc^{(1)} / dt = x_m c^{(0)} [-k_d + k_g m^{(0)}] + I x^* + (c_{in}^{(1)} - c^{(1)}) / \tau_r \quad (2.6)$$

$$\begin{aligned} dc^{(2)} / dt = & x_m^2 c^{(0)} [k_g m^{(0)} + k_d] + 2x_m c^{(1)} [-k_d + \\ & k_g m^{(0)}] - k_b c^{(2)} / 3 + 2k_a c^{(1)2} + I x^{*2} + (c_{in}^{(2)} - c^{(2)}) / \tau_r \quad (2.7) \end{aligned}$$

Equations 2.4 and 2.6 satisfy the mass balance, $x_m dm^{(0)} / dt = -dc^{(1)} / dt$, describing how mass transforms between the liquid phase or cluster (crystal or solid) phase. Thermodynamic equilibrium conditions in eqs 2.4–2.6 are determined by setting the time derivatives $d[] / dt = 0$, vessel residence time $\tau_r \rightarrow \infty$, and nucleation rate $I = 0$. The results are

$$m_{eq}^{(0)} = k_d / k_g \quad \text{and} \quad c_{eq}^{(0)} = k_b / k_a \quad (2.8)$$

Zeroth moments then are consistent with thermodynamic equilibrium. Substituting the above relations into eq 2.7 for the *second moment*, however, does not yield zero for the time derivative, because ripening behavior can still occur, as we will explain shortly.

Define the dimensionless quantities

$$\begin{aligned} C^{(n)} = & c^{(n)} / m_{eq}^{(0)} x_m^n, \quad \theta = t k_d, \quad \tau = \tau_r k_d, \\ \beta = & k_b / k_d = k_a c_{eq}^{(0)} / k_d, \quad S = m^{(0)} / m_{eq}^{(0)}, \\ J = & I^* / m_{eq}^{(0)} k_d, \quad x_s = x^* / x_m, \\ \alpha = & k_a m_{eq}^{(0)} / k_d = k_a / k_g \quad (2.9) \end{aligned}$$

All rate terms are scaled by the monomer dissociation term, k_d . Relating each rate term to k_d as a ratio allows a comparison of competitive rates of nucleation (J), cluster fragmentation or breakage (β), and aggregation (α). Differential equations (2.4)–(2.7), when divided by $x_m^n k_d m_{eq}^{(0)}$, yield the dimensionless differential equations

$$dS / d\theta = -(S - 1) C^{(0)} - J x_s + (S_i - S) / \tau \quad (2.10)$$

$$dC^{(0)} / d\theta = (\beta - \alpha C^{(0)}) C^{(0)} + J + (C_i^{(0)} - C^{(0)}) / \tau \quad (2.11)$$

$$dC^{(1)} / d\theta = (S - 1) C^{(0)} + J x_s + (C_i^{(1)} - C^{(1)}) / \tau \quad (2.12)$$

$$\begin{aligned} dC^{(2)} / d\theta = & (S + 1) C^{(0)} + 2(S - 1) C^{(1)} - \beta C^{(2)} / 3 + \\ & 2\alpha C^{(1)2} + J x_s^2 + (C_i^{(2)} - C^{(2)}) / \tau \quad (2.13) \end{aligned}$$

These scaled differential equations are nonlinear by the appearance of quadratic terms: $C^{(0)2}$, $SC^{(0)}$, $SC^{(1)}$, and $C^{(1)2}$. By setting $J = 0$, $1/\tau = 0$, and $d[] / d\theta = 0$, we determine the equilibrium conditions (eq 2.8) for the closed (no flow) case

$$S_{eq} = 1 \quad \text{and} \quad C_{eq}^{(0)} = \beta / \alpha \quad (2.14)$$

Some characteristic behavior can be deduced for the steady-state flow case, but the batch, no-flow case is of more immediate interest. Therefore, we turn to the *unsteady-state batch equations*, written by setting $1/\tau \rightarrow 0$.

$$dS / d\theta = -(S - 1) C^{(0)} - J x_s \quad (2.15)$$

$$dC^{(0)} / d\theta = (\beta - \alpha C^{(0)}) C^{(0)} + J \quad (2.16)$$

$$dC^{(1)} / d\theta = (S - 1) C^{(0)} + J x_s \quad (2.17)$$

$$\begin{aligned} dC^{(2)} / d\theta = & (S + 1) C^{(0)} + 2(S - 1) C^{(1)} - \beta C^{(2)} / 3 + \\ & 2\alpha C^{(1)2} + J x_s^2 \quad (2.18) \end{aligned}$$

Dynamics of Nucleation, Growth, and Aggregation

The above nonlinear ordinary differential equations (2.15)–(2.18) are straightforward to solve numerically. Simple cases are irreversible cluster aggregation ($\alpha > 0$, $\beta = 0$, and $S_0 = 1$) and irreversible cluster fragmentation ($\alpha = 0$, $\beta > 0$, and $S_0 = 1$). Setting $S_0 = 1$ specifies that there is no nucleation and no cluster growth. For cluster growth or dissolution, we set the initial supersaturation, S_0 , either greater than or less than 1, respectively, with $\alpha = 0$ and $\beta = 0$.

For Ostwald ripening, also called isothermal recrystallization,¹ larger clusters grow at the expense of smaller nuclei that denucleate by dissolving and releasing solute. Nielsen⁵ asserted that ripening must occur near saturation; otherwise, growth or dissolution would dominate the recrystallization. The present model al-

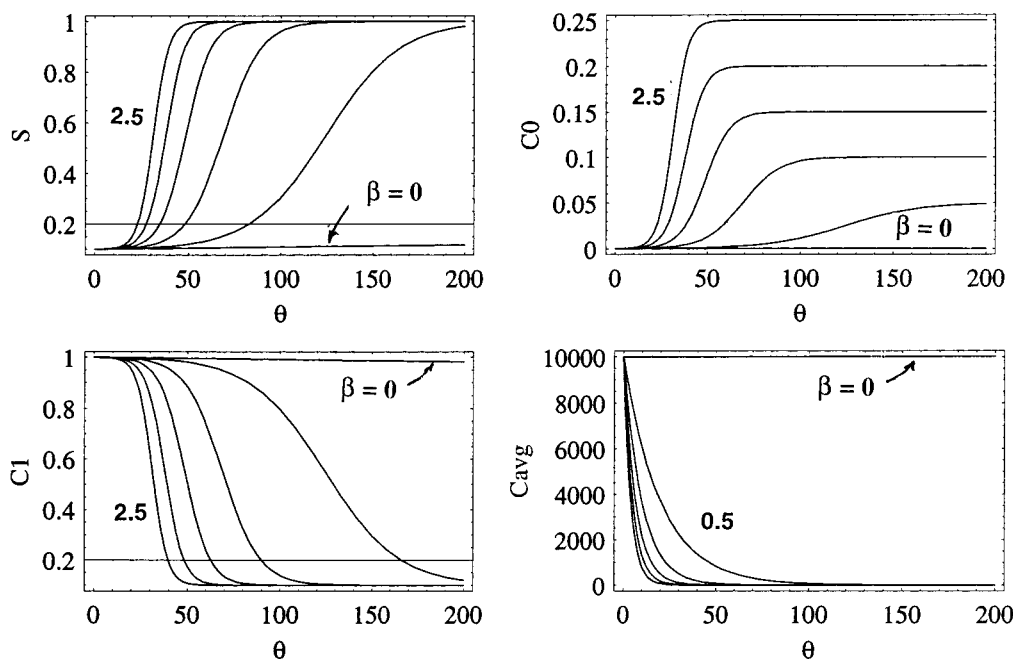


Figure 1. Cluster dissolution (melting) for $T = 473$, $C_0^{(0)} = 0.0001$, $C_0^{(1)} = 1$, $S_0 = 0.1$, $j = 0$, and $\alpha = 1$ with $\beta = 0-2.5$ in steps of 0.5 (right to left).

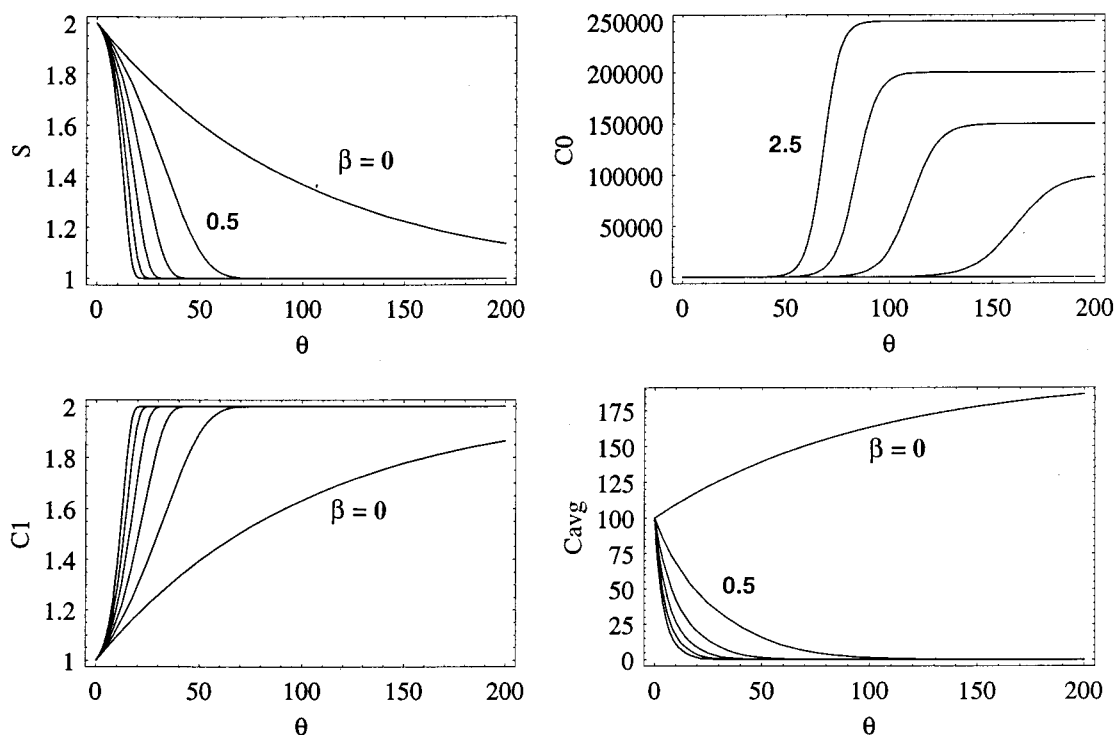


Figure 2. Cluster growth with breakage for $C_0^{(0)} = 0.01$, $C_0^{(1)} = 1$, $S_0 = 2$, $j = 0$, and $\alpha = 10^{-6}$ with $\beta = 0-2.5$ in steps of 0.5 (right to left). When $\beta = 0$, only cluster growth occurs.

lows a simple approximation for ripening if we set $S_0 = 1$ and change the sign on J in eq 2.16. This provides that nuclei of critical (minimum) size dissolve (denucleate) and their solute redeposits on larger clusters. Even for $S = 1$ the second moment continues to change by growth dispersion, which appears naturally in this reversible theory. A more complete analysis would incorporate the important effect of interfacial energy on solubility. The tendency for ripening to occur at $S = 1$ is an example of evolution of a metastable state. Such metastability has been mentioned²⁹ as a potential limitation of constructing nanostructures by chemical

self-assembly. The discussion was illustrated by experiments for nanocrystals growing by vapor deposition on surfaces, but the reasoning presented here suggests that the principle applies to crystal growth in general.

For cluster dissolution (melting) we set the nucleation rate equal to zero ($J = 0$), select an undersaturated condition ($S_0 < 1$), and choose C^{avg} to have a large value. For the conditions $\alpha = 1$, $S_0 = 0.1$, $C_0^{(0)} = 10^{-4}$, $C_0^{(1)} = 1.0$, i.e., $C^{avg} = 10^4$, we let β vary from 0 to 2.5. The results (Figure 1) show that supersaturation S rises as cluster mass $C^{(1)}$ falls. Because of the mass balance, eqs 2.15 and 2.17 are equal but opposite in sign, and thus

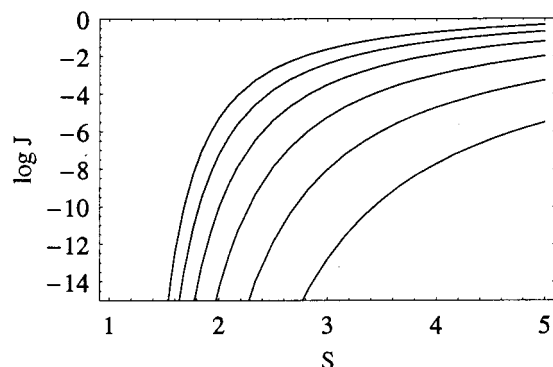


Figure 3. Dimensionless nucleation rate, J , versus supersaturation, S , at temperatures (right to left) 50, 100, 150, 200, 250, and 300 °C.

S and $C^{(1)}$ are mirror images. The supersaturation and number of clusters rise to their equilibrium values, $S_{eq} = 1$ and $C_{eq}(0) = \beta/\alpha$, and the cluster mass falls to zero. Decreasing the breakage rate (β) slows the dissolution. The number of clusters $C^{(0)}$ increases for $\beta > 0$ as clusters break up and C^{avg} falls precipitously. For $\beta = 0$ no cluster fragmentation occurs and the clusters dissolve by single-molecule dissolution. This slow process is reflected in the slight changes of the four variables plotted in Figure 1 for $\beta = 0$.

Cluster growth by monomer deposition is illustrated by calculations with $k_g \gg k_a$, for example, by $\alpha = k_a/k_g = 10^{-6}$. Figure 2 shows such cluster growth when fragmentation also occurs ($\beta > 0$). Again, the S and $C^{(1)}$ curves are mirror images. The case for $\beta = 0$, quite different owing to lack of fragmentation, displays the expected cluster mass growth with constant cluster number. Such cluster aggregation causes an increase in the viscosity for the supercooled (nonnucleated) liquid and represents vitrification if the viscosity reaches a

large value. Development of such amorphous structures can frustrate self-assembly processes.

For solidification by crystal nucleation, growth, and aggregation, the dimensionless nucleation rate is written as

$$J = j \exp(-E^*/RT) \quad (3.1)$$

where E^* is given by eq 1.1 and the dimensionless preexponential factor is $j = k^*/m^{(0)}_{eq}k_d$. Nucleation parameters chosen for the calculations are for amorphous silica:³ $v = 27.27 \text{ cm}^3/\text{mol}$ and $\sigma = 46 \text{ erg/cm}^2$. The dimensionless nucleation rate for amorphous silica as a function of supersaturation and temperature is shown in Figure 3. Reducing the preexponential factor j simply shifts the curves down. Clearly the different combinations of possible values of rate parameters encompass a huge time range. As Fourkas et al.³⁰ assert, supercooled liquids embrace extensive multiscale time behavior.

To illustrate the nucleation effect, we choose an initially supersaturated liquid with no crystals present ($C_0^{(0)} = 0$ and $C_0^{(1)} = 0$), except those of critical size being formed by nucleation. The initial conditions are $C_0^{avg} = 0$, $S_0 = 2$, $T = 730 \text{ K}$, $j = 6.4$, $\alpha = 0.2$, and $\beta \geq 0$. Nucleation retards the beginning of crystal growth as shown in Figure 4. The induction time is relatively independent of growth, aggregation, or deaggregation rates. Supersaturation can influence the behavior enormously. Consider the case of Figure 4, but with a slightly higher supersaturation, $S_0 = 3$. Figure 3 indicates the extreme increase in the nucleation rate. As shown in Figure 5, the induction lag is less evident, because the nucleation rate is so much increased relative to Figure 4. The number of crystals reaches a maximum (Figure 5B) and falls off as the influence of aggregation is realized.

If the temperature is reduced to 473 K, the nucleation

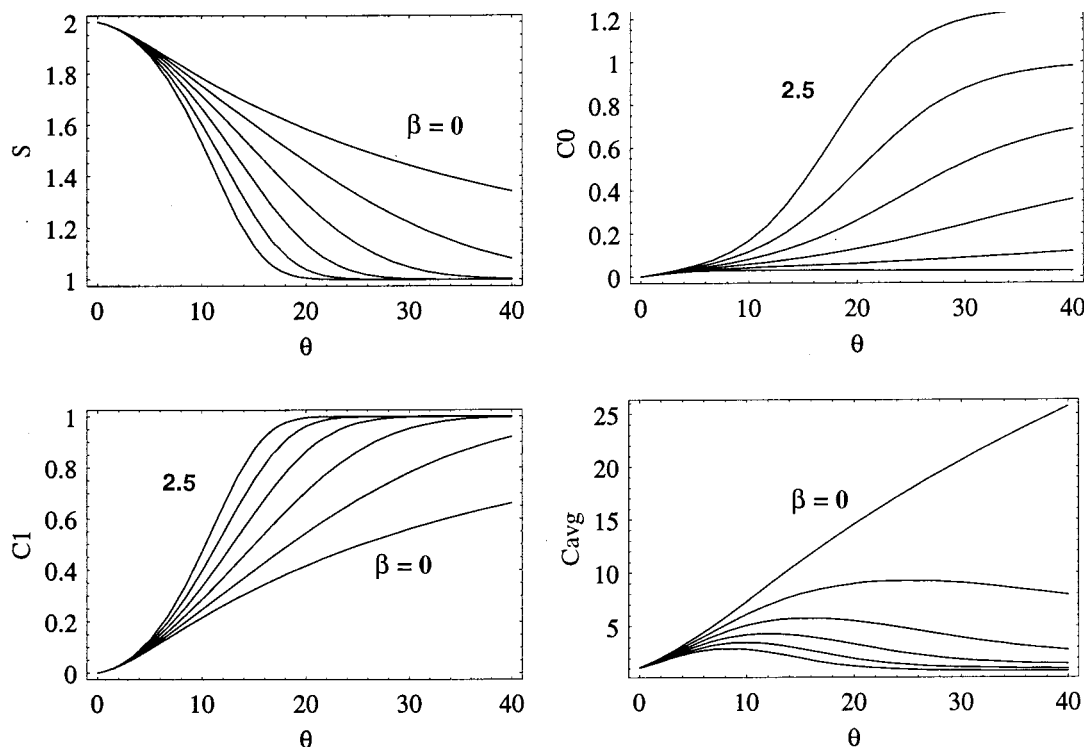


Figure 4. Nucleation induction in a liquid ($C_0^{avg} = 0$) for $S_0 = 2$, $j = 6.4$, $T = 730 \text{ K}$, and $\alpha = 0.2$ with $\beta = 0-2.5$ in steps of 0.5 (right to left).

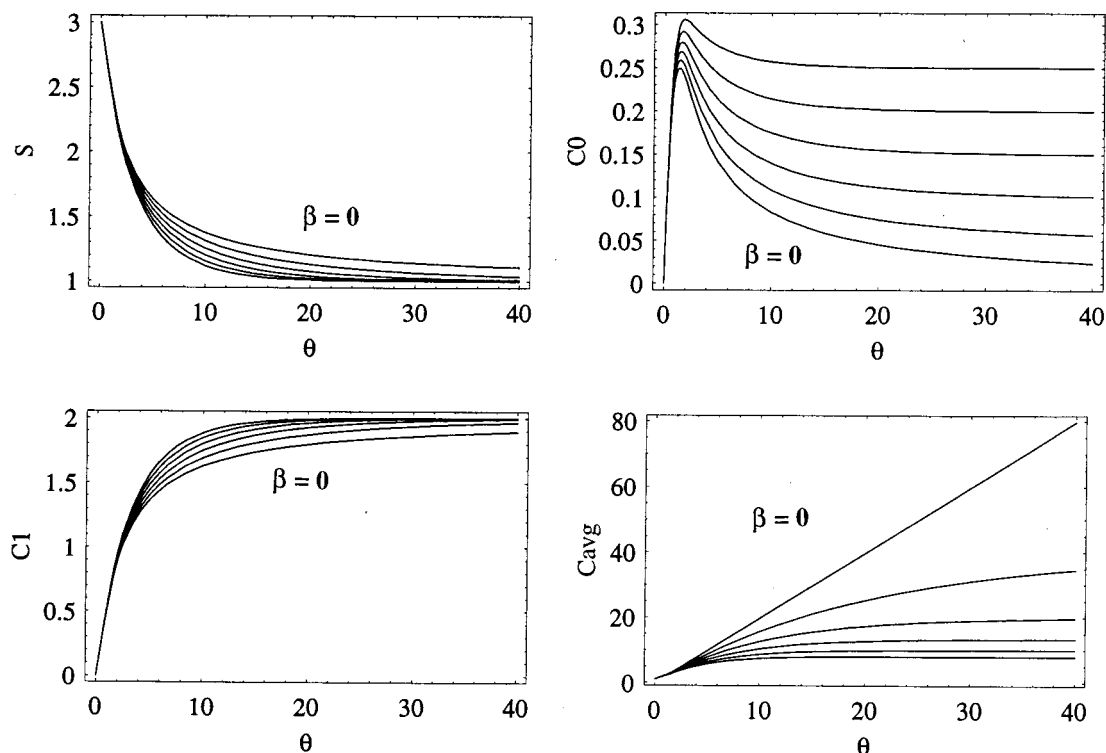


Figure 5. Cluster nucleation and growth in a liquid ($C_0^{\text{avg}} = 0$) for $S_0 = 3$, $j = 6.4$, $T = 730$ K, and $\alpha = 1$ with $\beta = 0-2.5$ in steps of 0.5 (right to left).

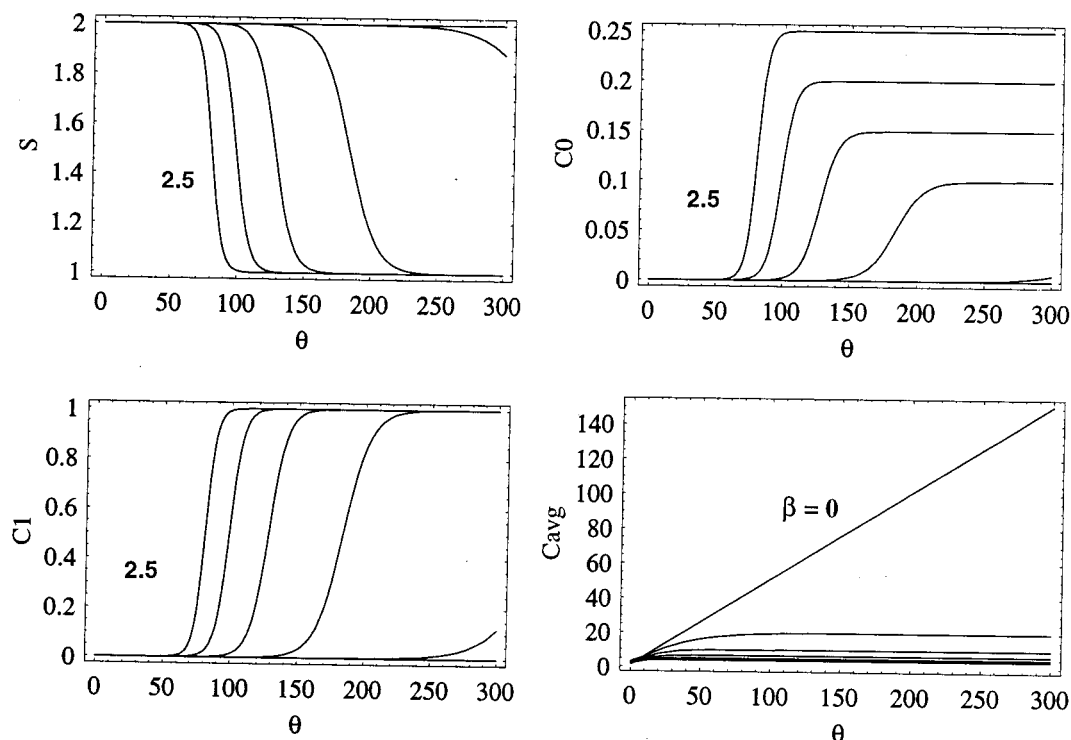


Figure 6. Cluster nucleation and growth in a liquid ($C_0^{\text{avg}} = 0$) for $S_0 = 2$, $j = 6.4$, $T = 473$ K, and $\alpha = 1$ with $\beta = 0-2.5$ in steps of 0.5 (right to left). The low temperature causes a large induction time preceding the decrease in S and appearance of observable clusters. Clusters start growing, however, at $t = 0$, as indicated by C^{avg} .

rate is greatly diminished. In Figure 6 we also change the aggregation and growth rate ratio to $\alpha = 1$. The induction time now is affected by the breakage rate as indicated in Figure 6. Clusters that fragment provide more surface area for growth and aggregation, resulting in increased mass and increased numbers (Figure 6B,C).

To summarize the calculated results, we see that all

of the closed systems approach equilibrium as time increases indefinitely. Time dependences for S and $C^{(1)}$ are mirror images of each other because of the mass balance. Simple nonnucleated processes, such as melting and cluster aggregation and growth, are reasonably characterized by the theory as are more complex competitive behavior among nucleation, growth, and ag-

gregation. Temperature and supersaturation have strong effects on the nucleation rates and hence on the CSD.

Conclusion

We have introduced a theoretical formalism that combines nucleation, growth, and aggregation. The model contains within it classical nucleation theory with reversible crystal growth (dissolution) and reversible cluster aggregation (deaggregation). Mass (population) balance equations for the CSD lead to dynamic moment equations for the molar and mass concentrations. Clusters are either present initially or nucleate by the classical free-energy activation barrier theory. Clusters much larger than nuclei can undergo fragmentation by binary cleavage or the reverse process, aggregation. The reversibility of growth and aggregation ensures that closed systems will evolve to equilibrium and that open systems with steady flow will evolve to steady state. Clusters can grow by monomer addition or dissolve by monomer dissociation. The ordinary differential equations for the moments are easily solved for a wide range of rate parameters and initial conditions, yielding various phase transformations. Results of computations with different nucleation rates show the competition among cluster nucleation, growth, and aggregation. Large nucleation rates yield large numbers of clusters that grow and aggregate. Small nucleation rates provide few new clusters, allowing growth from the supersaturated liquid and reversible aggregation. For a large temperature decline, the average cluster mass increases strongly by aggregation, causing a huge viscosity increase and vitrification. Aggregation overwhelming crystal growth leads to glassy, supercooled liquids. The phase transition behaviors that can be described by this kinetics approach are as follows: (i) cluster growth by monomer addition ($S > 1$), (ii) cluster dissolution by monomer dissociation (melting), (iii) classical nucleation by monomer addition, (iv) denucleation by monomer dissociation ($S < 1$), (v) Ostwald ripening (coarsening) for $S \sim 1$, (vi) induction time for precipitation caused by a low nucleation rate, (vii) glass transition (vitrification) due to cluster aggregation with a low nucleation rate.

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